

Project Work Proposal Summer Semester 2026

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DriveAlert

A Camera-Based Driver Drowsiness Detection and Multi-Modal Alert System

System Overview

In this section, we describe the DriveAlert system that we have chosen for our ISE II project. We explain what the system does and why we chose it (1.1), what hardware and software it is made of (1.2), how it is supposed to behave in different situations (1.3), and we include some early diagrams to support our idea (1.4).

1.1 General Information

For our ISE II project, we decided to work on a system called DriveAlert — a camera-based driver drowsiness detection and alert system. The idea behind this system is simple: the car watches the driver and warns them if they are getting too tired to drive safely.

We chose this topic because driver fatigue is a very real and serious problem. Many road accidents happen not because of a mechanical failure, but because the driver fell asleep or lost focus. According to the World Health Organization, around 1.19 million people die in road accidents every year, and drowsiness is one of the main contributing factors — especially on highways and during long night drives. We thought it would be interesting and meaningful to model a system that could actually help prevent some of these accidents.

Another reason we picked this topic is that Driver Drowsiness and Attention Warning (DDAW) systems are now actually required by EU regulations for new cars. Companies like Bosch, Continental, and even HELLA/Forvia — which is based right here in Lippstadt — already make such systems. So it is not just a theoretical idea, it is a real product that is being developed and sold today.

DriveAlert works by placing an infrared camera inside the car, pointing at the driver's face. The camera runs 24/7 and sends video to a small ECU (computer) that analyzes what it sees. The ECU looks for signs of drowsiness such as:

- **PERCLOS:** the percentage of time the eyes are closed. If the eyes are closed more than 15% of the time, the driver is probably getting tired.
- **Blink rate and duration:** longer and more frequent blinks are a sign of fatigue.
- **Yawning:** detected by the mouth opening wide.
- **Head nodding:** the head slowly dropping forward, which indicates micro-sleep.

Based on these cues, the system classifies the driver into one of three states: Awake, Mildly Drowsy, or Severely Drowsy. Depending on the state, it sends the right type of alert to get the driver's attention — starting with a small visual warning and escalating to sound and vibration if needed. In the worst case, it also shows where the nearest rest stop is on the navigation screen.

The system is designed for standard passenger cars and is especially useful on highways and long trips. The driver is always in full control — DriveAlert only observes and warns, it does not take over the car.

1.2 System Specifications

DriveAlert is made up of an in-cabin camera as the main sensor, plus some secondary inputs from the car itself (like the steering angle), an ECU that does all the processing, and a set of output devices to alert the driver. Table 1 below lists all the main components with their details.

DriveAlert System Specifications

Component	Description	Used For
IR Driver Camera	An infrared 1080p camera (30 fps) that is mounted on the steering column and can see the driver clearly even at night thanks to near-IR lighting.	Drowsiness detection
Steering / Lane Input	The system also reads the steering angle and lane position from the car's existing sensors. These are used as extra signs of fatigue — for example, if the driver starts drifting.	Secondary confirmation
Drive-Time Monitor	A real-time clock that tracks how long the driver has been driving without a break. The longer the drive, the higher the risk of fatigue.	Fatigue risk weighting
Driver Monitoring ECU	The main computer of the system. It runs a real-time OS and processes the camera images every 50 ms. It runs the fatigue detection algorithm and decides which alerts to send.	All functions
Dashboard Display	Shows a warning icon and a short text message on the instrument cluster when drowsiness is detected.	Stage 1 — visual alert
Buzzer / Chime	Plays a beeping sound if the visual warning is not enough. The sound gets louder as drowsiness gets worse.	Stage 2 — auditory alert

Haptic Vibration	A vibration motor in the steering wheel or driver seat that activates when the situation becomes serious.	Stage 3 — haptic alert
Navigation Link	When the driver reaches the severe drowsiness state, the system shows the nearest rest stop on the navigation display so the driver knows where to pull over.	Stage 3 — rest advice
Speaker / Voice Output (TTS)	An in-cabin speaker with text-to-speech. When severe drowsiness is detected, the system speaks directly to the driver and suggests an action — for example, announcing the nearest rest area.	Stage 3 — voice guidance
Telematics Unit + GPS	A connected module with GPS positioning. If the driver becomes unresponsive after the haptic alert, it places an automatic emergency call (eCall) and transmits the vehicle's location.	Stage 4 — emergency SOS
Exterior Lights & Horn Link	A link to the car's exterior lighting and horn control unit. When the driver is unresponsive, the system flashes the hazard lights (and may sound the horn) to warn surrounding vehicles that this car has a problem.	Stage 4 — warn other cars
CAN Bus / Ethernet	CAN bus (500 kbit/s) is used for control signals between the ECU and the car. Automotive Ethernet is used to send the camera video to the ECU (too much data for CAN).	System communication
Power Supply	Standard 12 V car battery. The ECU has watchdog timers and voltage protection to make sure it does not fail or restart unexpectedly.	System reliability
Operating Conditions	The camera can handle different lighting, glasses, varied skin tones, and partial face occlusion. It is built to handle road vibration (ISO 16750 standard).	Environmental robustness
Software	The software is written in C/C++ using OpenCV for image processing and a machine learning classifier for fatigue detection. It follows the AUTOSAR layered software model.	All functions

One thing we think is important to mention about our system design is how the two sensing channels work together. The camera is the primary sensor — it detects drowsiness by looking at the driver's face. But the steering and lane signals from the car are used as a secondary check. For example, if the camera detects a long blink at the same time as the car starts to drift slightly in the lane, the system is much more confident that the driver is actually tired. This reduces false alarms (like triggering on a single long blink from someone who just rubbed their eye) and also makes the system more reliable when the camera view is partially blocked

1.3 Expected Behavior

Below we describe how DriveAlert is expected to behave in different driving situations. The system runs in the background the whole time the car is on, going through different states depending on what the camera and sensors detect.

Normal Driving — Everything is Fine

When the driver first starts the car, the system runs a quick self-test to check that the camera and all other components are working. If everything is OK, it goes into Monitoring mode. In this mode, the camera is running and the ECU is analyzing the video, but the driver is alert (PERCLOS below 15%, no yawning, no head nodding), so nothing happens. The driver does not notice the system at all — it is just quietly watching in the background.

Situation 1 — The Driver is Getting a Little Tired (Mild Drowsiness)

If the system detects that PERCLOS rises above 15% or the blink rate increases noticeably, it enters the Mild Drowsiness state. At this point, the driver is getting tired but is not yet in immediate danger. The system starts with a gentle warning:

1. **Stage 1 — Visual alert:** an amber icon and a short message like "Take a break soon" appears on the dashboard display.
2. **Stage 2 — Auditory alert:** if the signs of drowsiness continue after the visual warning, a chime or beep plays to make sure the driver notices.

If the driver wakes up — maybe they opened a window, turned up the radio, or just reacted to the alert — and the drowsiness indicators go back below the threshold, the system returns to normal Monitoring mode. No further action is needed.

Situation 2 — The Driver is Very Tired (Severe Drowsiness)

If the drowsiness keeps getting worse — PERCLOS reaches 30% or above, or a clear head-nod is detected — the system enters Severe Drowsiness state. At this point, the driver is in real danger of falling asleep. The system escalates the alerts significantly:

1. **Stage 3 — Haptic alert:** the steering wheel or driver seat starts vibrating to physically wake the driver up.
2. **Rest stop advice:** at the same time, the navigation display shows where the nearest rest area or motorway service station is, so the driver knows where they can safely stop.
3. **Voice guidance:** the system also speaks directly to the driver through the cabin speaker — for example "You seem very tired, the nearest rest area is 3 km ahead" — so the driver knows exactly what to do.

The system keeps the alerts going until the driver clearly responds — for example by pressing a button on the steering wheel to confirm they are awake, or by changing their posture and opening their eyes. Once the drowsiness indicators drop back to normal, the alerts stop and the system returns to Monitoring. If the driver dismisses the alert but then gets drowsy again shortly after, the system will re-alert. However, if the driver does not respond to the severe alert at all, the system escalates to the Emergency state (Situation 3).

Situation 3 — The Driver Does Not Respond (Emergency / Unresponsive)

If the severe alerts (haptic, voice, and rest advice) do not get any reaction from the driver — the driver does not press the steering-wheel button, does not change posture, and the drowsiness indicators stay critical for a short timeout (for example 10 seconds) — the system assumes the driver may have fallen asleep or become unresponsive, and enters the Emergency state. This is the most serious state, and the system now acts to protect both the driver and other road users:

1. **Automatic SOS call:** the system places an automatic emergency call (similar to the EU eCall system) through the telematics unit and transmits the vehicle's GPS location, so help can be sent if needed.
2. **Warning to other vehicles:** the system flashes the hazard lights (and may sound the horn) to warn surrounding traffic that this car has a problem, so other drivers can keep a safe distance.
3. **Continued voice and alerts:** the haptic and voice alerts keep running, still trying to wake the driver.

If the driver responds during this state — by pressing the button or clearly becoming alert again — the system cancels the emergency call (within a short grace period so a false alarm is not sent), stops the hazard signal, and returns to Monitoring. The driver always stays in full control; even in the Emergency state, DriveAlert never touches the steering, brakes, or engine.

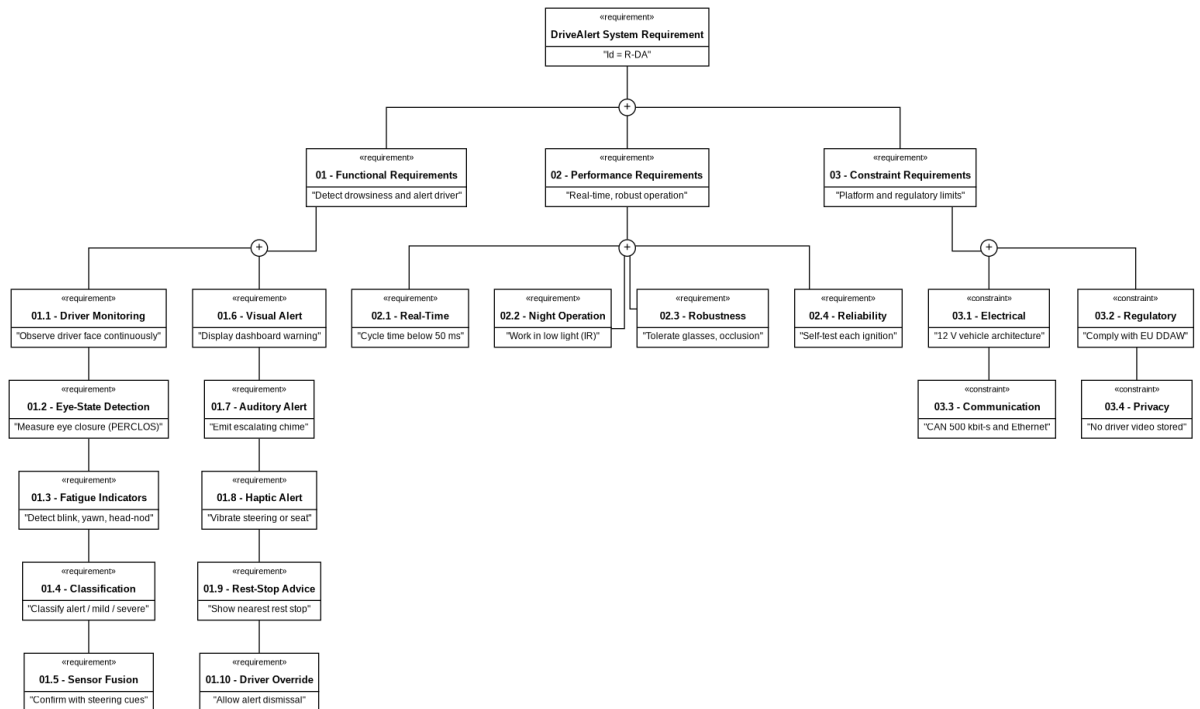
Situation 4 — Something Goes Wrong (Degraded Mode)

If the camera stops working (for example, if it is blocked or there is a hardware problem), or if the ECU detects a software error, the system immediately enters Degraded mode. In this state, it is impossible to know if the driver is drowsy, so the system does the safe thing: it turns off all automatic alerts (since it cannot rely on them anymore) and shows a permanent fault warning icon on the dashboard so the driver knows the system is not working. It also saves an error code (DTC) to the car's memory so a mechanic can find out what went wrong. The system stays in this state until the car is turned off and on again, and the self-test passes successfully on startup.

In all situations, the driver is completely in control. DriveAlert only observes and warns — it does not control the steering, brakes, or engine. The driver can always choose to ignore the alerts (though we hope they would not!).

1.4 Supporting Figures and Diagrams

Figure 1 shows the system requirements diagram, the basis for the SysML requirements package. The top-level system requirement (R-DA) is decomposed into functional, performance, and constraint requirements, each refined into specific sub-requirements.



As part of this project work, we are going to develop all four SysML model packages. For the Requirements Model, we will create a Requirements Diagram. For the Structural Model, we will develop a Block Definition Diagram (BDD) and an Internal Block Diagram (IBD). For the Behavioral Model, we will build a Use Case Diagram, an Activity Diagram, a State Machine Diagram, and a Sequence Diagram. Finally, for the Parametric Model, we will develop a Parametric Diagram. All these diagrams will be created using a SysML modeling tool and submitted as part of the final project.